Michael Rubin MEng Mechanical Engineer
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Profile Summary

Driven MechE graduate with hands-on experience prototyping, testing, and iterating real-world mechanical systems. Motivated to apply my technical depth and creative problem-solving in a ***design or manufacturing role****. With projects taken from analysis to functional hardware under tight timelines, I bring a strong background in* ***engineering analysis, rapid prototyping and design optimization****. Fluent in* ***5 languages****, I collaborate effectively across multidisciplinary teams in fast-paced environments.*

Experience

Mechanical Engineer | *MPC Lab, UC Berkeley – Berkeley, CA, USA* — *June – Present*
- Contract role supporting NIWC Pacific on development of an **all-terrain autonomous surveillance rover** (Project LUCI).
- Redesigning 3D-printed chassis of the rover to **improve structural robustness and usability**.
- Developed and implemented **testing protocols** to validate the control stack for autonomous navigation and localization.

Chassis Engineer - Driver Safety Lead| *CALSOL Solar Vehicle Team – Berkeley, CA, USA* — *Sept 2025 – Present*
- Responsible for GenXI seatbelt implementation in the occupant cell, ensuring ASC and WSC **regulation compliance**.
- Simplified shoulder belt anchorage, enabling in-house manufacturing. **Reduced weight by 40%** using topology optimization.
- Designed and executed **Instron validation tests** ensuring mounts withstand crash load with 1.6 FOS.
- Managed large CAD assemblies, prepared drawing packages using **GD&T** and collaborated with suppliers.

AutoCAD Drawing Engineer | *KBC Acurity – Remote* — *Oct 2022 – June 2025*
- Produced over 300 **technical drawings** for nationwide security camera installations at KBC Bank offices.

Teaching Assistant (UGSI: Statics, Mechanics of Materials) | *TU Delft – Delft, ZH, The Netherlands* — *Sept 2022 – Jan 2023*

Projects

Handheld Autorefraction Device | *UC Berkeley - Continued Independently* — *Jan 2025 – Present*
- **Led optical design** of a Scheiner disk based vision screening system to ± 0.50D SPH accuracy. Secured $1K grant for prototyping.
- Created alignment jigs using SolidWorks and developed **calibration protocols** to verify and maintain system accuracy.
- Supported development of an image-processing pipeline to compute corrective lens power.

Tactile Robotic End Effector, Graduate Capstone | *Embodied Dexterity Group – UC Berkeley* — *Sept 2025 – May 2026*
- Scaled a multi-chamber suction based system from a research-grade prototype to a **production-ready unit for 1,000+ orders**.
- Standardized the robot-arm interface, reduced part count 32→9 and cut setup time 15min→1:45min.
- Applied **design-for-manufacturability principles and FMEA** to redesign injection-molded parts for better serviceability and reliability.
- Designed and integrated custom PCB with BOM management using KiCad while ensuring clean cable routing.

Light module for Robotic Surgery System, BSc Thesis | *AdLap Lab – TU Delft* — *Jan 2024 – June 2024*
- Designed **operating room compliant** detachable light module for a low-cost laparoscopic system, with white/IR mode switching.
- Achieved 3x illuminance while reducing device size by 50% relative to prior prototype.
- Implemented mechanically actuated **mode switching with precise alignment**, guided by 2D ray tracing and optical bench tests.
- Engineered thermal management system using **heat transfer analysis** and validating performance via long-duration thermal testing.

CNC machine | *TU Delft* — *Sept 2023 – Dec 2023*
- Designed and built a CNC machine for PMMA carving using an H-bot single-belt gantry architecture;
- Cut footprint by 10% vs benchmark 3-axis system and minimized part count. Placed top 5% in faculty competition.
- Integrated belt pre-tensioning and vibration damping to improve **motion control and reliability**.

Relevant Skills

Mechanical: CAD & Drawing (SolidWorks CSWA certified), FMEA, Tolerancing, GD&T (ASME Y14.5-2018), DFx, RCA, V&V, Material Selection (Granta), BOM Management, Motor & Sensor Selection, Thermal Optimization, Controls (PID, MPC), Assembly Modeling

Manufacturing: CNC, Additive (FFF, Cura), Injection Molding, Sheet Metal Fabrication, Bench Tests, Machine Shop Trained

Software: FEA (Abaqus, Ansys, COMSOL), Python, MATLAB, C++, G-Code, Git, Ros2, VSCode, Linux, Excel (PivotTables)

Languages & Soft Skills: English (near-native, TOEFL 114/120), French (native), Dutch (native), Spanish (fluent), Hebrew (basic), Reporting & Documentation, Proactive, Adaptability, Collaboration, Task Prioritization, Creativity

Education

University of California at Berkeley (UC Berkeley) – GPA 3.8/4 — **Berkeley, CA, USA**
Master's degree in Mechanical Engineering – Biomechanics & Control — *Aug 2025 – May 2026*

Delft University of Technology (TU Delft) – GPA 3.8/4 — **Delft, The Netherlands**
Honours Bachelor's degree in Mechanical Engineering — *Aug 2021 – June 2024*
- Exchange semester at Norwegian University of Science and Technology (NTNU) – GPA 3.825/4

Awards & Honors

Graduate Scholarship | *Belgian American Education Foundation* — *Aug 2025 – May 2026*
One of 96 Belgians awarded a national grant to pursue graduate studies in the United States; covering full tuition and stipend.

Honours Program Delft Certificate | *TU Delft* — *Sept 2022 – June 2024*
Selected as 1 of 4 MechE students for exclusive program; completed 20 additional credits within nominal graduation time.